Mathematical review

Learning Manifolds from Discrete Data
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I found several definite errors and several substantial gaps. The original population recovery argument in Section 3 appears largely sound. The proposed replacement of its rigidity energy is not equivalent, and parts of the finite-sample regularity argument require new hypotheses or replacement proofs.

The most useful repair is short: the reach estimate in Proposition 4.9 follows directly from Taylor expansion of the retraction. This avoids its incorrect global derivative estimate and its use of curvature in place of reach. A second simplification makes Lemma 4.20 deterministic.

Priority	Location in draft	Finding
High	p. 5; Observation	The proposed E_2' has smooth zero-energy nonprojection minimizers.
High	pp. 19-20; Prop. 4.9	Equation (17) is false, even for the exact projection. The reach conclusion has a replacement proof.
High	pp. 26-32; Section 4.2	Uniform regularity, a common projection domain, and the slicing argument need explicit hypotheses and proof.
High	p. 32; Cor. 4.29	The final unconditional tangent claim contradicts Remark 4.4.
Medium	pp. 18-19; Lem. 4.8	False for incomplete submanifolds; the curve in its proof must be a geodesic.
Medium	pp. 21-25, 30-32	Domain, topology, sampling, scaling, and interpolation errors are listed below.

Conventions. Page numbers below are the printed PDF page numbers. I write $A = M_n$, $\tau = \mathrm{reach}(M)$, b for the containment slack (the underlined δ in the paper), and s for Sobolev order (the paper also uses k for codimension). A finding labelled gap does not assert that the conclusion is false. Proposed repairs are distinguished from results already proved in the draft.

1. E_2' cannot replace E_2 in the recovery theorem

p. 5, Observation | Explicit counterexample to equivalence of the two energies

Let M be a circle of radius R, with $0 < \delta < 2R/7$. Write a point in its tube as $x = (R+t)u$, where $|u| = 1$ and $-\delta \le t \le \delta$. Choose a nonzero smooth bump a supported in $(\delta/2, 3\delta/4)$, with $0 \le a \le \delta/8$, and define

$$T_a((R+t)u) = (R + a(t))u.$$

This map fixes M, its image lies within $\delta/8$ of M, and its displacement is $|t-a(t)| \le \delta$. Hence E_1, E_3, and E_4 all vanish. Every a(t) lies in $[0, \delta/8]$, where a and its derivative vanish. The radial derivative of T at T(x) is therefore zero, while T(x)-x is radial. Consequently

$$DT_a(T_a(x))(T_a(x) - x) = 0, \qquad E_2'(T_a) = 0.$$

Nevertheless T_a differs from π_M wherever a is nonzero, and is not idempotent: $T_a(T_a(x)) = Ru$. This example is smooth up to the tube boundary and meets the population hypotheses. It does not refute Theorem 3.1, which uses the original E_2.

Fix. Retain E_2 for the recovery theorem. If E_2' is used in the algorithm, describe it as a distinct objective without the current recovery guarantee. Keeping E_2 as an additional term restores the existing argument. Adding an idempotence penalty would rule out this counterexample, but would still require a fresh consistency proof for the modified objective.

2. Equation (17) is false away from the image

pp. 19-20, Proposition 4.9 | Definite error; reach conclusion repairable

At $y \in A$, DT(y) is an oblique projection, so its norm is the secant of the angle between the fiber and the normal of A. At a point outside A, DT need not be a projection. Equation (17) identifies the supremum on the whole tube with the norm at image points.

For $T = \pi_M$ and a circle of radius R, the fibers are exactly normal and A = M, so the angle θ in that equation is zero. But on the inner side of the tube,

$$\|D\pi_M\|_{L^\infty(M_{\delta'})} = \frac{R}{R - \delta'} > 1.$$

For fixed $\delta/R > 0$ this does not approach 1 as h tends to zero, whereas (17) claims $1 + O(h^2/\delta^2)$. Choosing R sufficiently large also meets any prescribed small positive Hessian allowance c_2/δ. Thus this is a counterexample within the intended setting, not merely for a general retraction.

There is a second gap on p. 20: bounded second fundamental form does not imply that the whole intersection of A with a Euclidean ball is a single graph. Nearby sheets can be connected elsewhere while curvature stays bounded. Likewise Federer's chord inequality with curvature K in the place of inverse reach is not available from curvature alone.

Replacement proof of Proposition 4.9

Preserves the stated lower bound; no estimate of the full derivative norm is needed

Let $U = M_{\delta'}$, $\varepsilon = \sup_{y\in A} d(y,M)$, $R_0 = \delta'-\varepsilon > 0$, and $K = \|D^2T\|_{L^\infty(U)}$, using the bilinear operator norm. The R_0-neighborhood of A is contained in U. Since T fixes A, DT(y) has image T_yA at $y \in A$.

Take $y_1,y_2 \in A$ and put $v = y_2-y_1$, $\ell = |v|$. If $\ell < 2R_0$, their entire chord is in U: every chord point is within $\ell/2$ of one endpoint. Taylor expansion and orthogonal projection onto $N_{y_1}A$ give

$$\|P_{N_{y_1}A} v\| \le \tfrac{K}{2}|v|^2.$$

If $\ell \ge 2R_0$, the trivial bound $\|P_N v\| \le |v|$ gives $\|P_N v\| \le |v|^2/(2R_0)$. Thus for every pair of points of A,

$$\|P_{N_{y_1}A}(y_2 - y_1)\| \le \tfrac{1}{2}\max\{K, R_0^{-1}\}|y_2 - y_1|^2.$$

Federer's reach characterization (1959, Theorem 4.18; reference [1]) now yields

$$\operatorname{reach}(A) \ge \min\{K^{-1}, R_0\}.$$

Taking $K = c_2/\delta$ gives precisely the lower bound in Proposition 4.9. This proof uses control of the full Hessian on the chord, which is stronger than control of the second fundamental form only.

3. Lemma 4.8 is false without completeness

pp. 18-19 | Missing hypothesis and invalid choice of curve

Take M to be the unit circle, and take $A = \{(1+t,0): |t| < \varepsilon\}$ for $0 < \varepsilon < 1/4$. This is a smooth submanifold without boundary as an abstract manifold, its second fundamental form is zero, and $\sup_A d(\cdot,M) = \varepsilon$. Its tangent is perpendicular to the circle tangent at every projected point. The left side of the asserted inequality is 1 and the right side is $2\sqrt{\varepsilon} < 1$.

The proof optimizes over a time that need not exist on an incomplete A. Also, a generic unit-speed curve in A can have arbitrarily large tangential acceleration: a second fundamental form bound controls the acceleration of a unit-speed *geodesic*, not every unit-speed curve.

Fix. Require A to be complete without boundary (compact suffices in the application), and choose a geodesic. The needed order can be obtained with, for example, the following safe constant:

$$\sin\angle(T_yA, T_pM) \le 5\sqrt{\varepsilon(K + \tau^{-1})}, \qquad \|\mathrm{II}_A\| \le K.$$

Here is a proof of this corrected version. Let c be the unit-speed geodesic with initial velocity u, put $q = \pi_M(c(t))$, and set $\sigma = |P_{NpM}u|$. Geodesic acceleration is bounded by K, $|q-p| \le t+2\varepsilon$, and Federer's tangency inequality on M gives

$$t\sigma \le 2\varepsilon + \frac{Kt^2}{2} + \frac{(t+2\varepsilon)^2}{2\tau}.$$

For $\varepsilon > 0$ choose $t = 2\sqrt{(\varepsilon/(K+1/\tau))}$; completeness permits this choice. Division by t gives $\sigma \le [2+2\sqrt{(\varepsilon/\tau)}+\varepsilon/\tau]\sqrt{(\varepsilon(K+1/\tau))} \le 5\sqrt{(\varepsilon(K+1/\tau))}$. The $\varepsilon = 0$ case follows by differentiation of $A \subset M$. This preserves the $O(h/\delta)$ application in (16), with a changed universal constant.

4. Energy minimization does not select Sobolev regularity

p. 6; Theorem 4.23 and Lemma 4.26, pp. 26-31 | Essential hypotheses need to be stated

The proof of Theorem 4.23 says that T_n minimizes its H^s norm. Its statement only says "a minimizer T_n"; elsewhere this means an arbitrary minimizer of E_n. These are different classes. Remarks 4.4 and 4.6 already show that zero energy permits arbitrary oscillation of a normal graph between sample points.

The distinction can be made completely explicit on a circle. Put $T(x) = p+g(p)$, $p = \pi_M(x)$, with g a normal section vanishing at the samples. On this circle, if $r(p) = d(p,M_n)$ and $x = p+t\nu(p)$, then

$$d(x, M_n)^2 = t^2 + (1 + t/R)r(p)^2.$$

Thus every x in the sample tube has δ-$|t| \geq (1-\delta/R)r(p)^2/(2\delta)$. Choosing $|g(p)|$ below a fixed small multiple of $r(p)^2/\delta$ ensures $E_4 = 0$. For example, take $b = 2h \leq \delta/8$ and $|g| \leq h$, so that $d(g(p)+p,M_n) \leq 2h$ and $E_3 = 0$ as well; E_1 and E_2 vanish automatically. A bump of height comparable to h^2/δ in an empty gap therefore remains an exact minimizer, and it may oscillate on an arbitrarily short wavelength.

Its H^s norm and curvature need not be uniformly bounded. Moreover, for $d = 1$ and $s \geq 5$, the displayed rate in Theorem 4.23 is faster than h^2 for fixed δ, so it cannot hold uniformly over all these energy minimizers with constants depending only on the displayed fixed quantities.

Fix. Define the selected maps explicitly in every regularity theorem. For example, minimize H^s norm over the zero-energy set with a specified Hessian bound $K \geq \|D^2\pi_M\|\infty$, and assume or prove existence in an appropriate closed function space. Alternatively state the results directly for maps with $E_n = 0$ and uniform H^s and Hessian bounds. Minimizing the *sum* proposed on p. 6 gives a bound by the corresponding sum of the two norms of π_M; it does not imply individual H^s minimality.

The alternating neural-network procedure on p. 6 is not a convergence proof for this constrained problem. Feasibility of π_M in a smooth function class does not prove that it is exactly representable by a given network or that the optimization reaches a global constrained minimizer.

5. The trace/slicing proof needs a geometric rewrite

pp. 28-30, Lemma 4.26 | Substantial gap; potentially repairable

A connected compact manifold cannot globally be flattened to $\mathbb{R}^d$. A tubular neighborhood is a normal disk bundle and need not be a global product $M \times B^{D-d}$. Hence the globally indexed parallel leaves M_z, and the global degree argument for g_z on each such leaf, are not justified under the stated assumptions.

Even in a trivial bundle, a nonlinear flattening does not preserve Euclidean orthogonality or straight fibers. The equation $x = y + Df(y)^t f(y)$ on p. 30 is the nearest-point equation for a graph over a *flat Euclidean plane*; curvature and metric coefficients must be retained for a graph over a curved M.

There is also a quantitative gap: transversality of ker DT to a tangent plane implies invertibility of a restriction of DT, but not uniform upper and lower singular-value bounds away from the image. A map can have the same kernel and image with very different nonzero singular values. The

incorrect full-tube use of the projection norm in (17) cannot supply these bounds.

Repair route. Work in finitely many normal-bundle charts with margins between smaller and larger patches. First prove uniform derivative and inverse bounds on a specified smaller tube. Then use Fubini to choose good transverse slices locally, use local quantitative inverse/composition estimates, and patch the resulting graph estimates with a partition of unity. Obtain the regularity of the nearest projection from the actual metric nearest-point equation by an implicit inverse argument. The constants must record the chart, inverse, and tube bounds.

The lemma is intended to avoid the ordinary codimension-dependent Sobolev trace loss. That improvement may be possible using the affine-fiber structure, but the bundle and quantitative inverse steps above are part of the proof, not a consequence of the standard trace theorem. At the last step on p. 30, the inverse needed in the composition is $\tilde{g}$, whereas the text names $\tilde{g}^{-1}$.

6. A common domain for the two orthogonal projections is missing

pp. 26-31, Proposition 4.25 and Section 4.2 | Expressions need not be defined under unconditional hypotheses

The estimates compare π_A and π_M on $M_{n,\delta}$. Remarks 4.4 and 4.6 permit reach(A) to be arbitrarily small. Consequently π_A is not guaranteed to be single-valued on that domain. Even Proposition 4.9 supplies only reach(A) ≥ min(1/K, δ′-ε), which need not cover the full δ sample tube.

Fix. After imposing the regularity hypotheses and using the corrected reach proof, choose an analysis radius with max(h,b) < r < min(1/K, δ′-ε), within the learned domain. State assumptions ensuring that such an r exists; for example Kδ ≤ 1, small h/δ, and b ≤ δ/8 permit r = δ/4. Because each X_i belongs to A, d(x,A) < r for $x \in M_{n,r}$; both projections are then defined there, and this domain contains M and A. Use the same r throughout the later estimates.

Proposition 4.25 also starts by asserting that every normal line of A meets M uniquely in a suitable point s(x). That requires a separate normal-projection/transversality argument on this smaller tube. It does not follow directly from $\pi_M|_A$ being a diffeomorphism. Tangent variation of A and derivative bounds for π_A bring in the reach/curvature of A, so these dependencies must be included in the constant.

On p. 27, a full ambient derivative norm is used where the statement only controls the derivative restricted to TM. These are not identical quantities. A projection-angle comparison and the off-surface shape-operator error are needed to pass between them.

7. Theorem 4.21 uses reach estimates that it has not proved

pp. 24-25 | Several proof gaps; the order Kh^2 can be retained conditionally

In Step 2 the tangency estimate is applied to A with 1/reach(A) replaced by $K = \|D^2T\|_\infty$. Step 1 only establishes $\|II_A\| \le K$. That alone is not the reach estimate needed for this substitution. The assumptions also revert to τ ≥ 5δ although Proposition 4.3 and Theorem 4.7 use τ ≥ 10δ and the stronger slack assumptions.

Fix for Step 2. For a nearby sample X_j and a sample X_i, Taylor-expand T along their chord inside the sample ball. Since both samples are fixed,

$$(I - DT(X_i))(X_j - X_i) = O(K|X_j - X_i|^2).$$

Projecting normally to $T_{X_i}A$ gives the chord estimate actually needed for the thick-simplex argument. This uses the full Hessian bound directly, without assuming reach(A) ≥ 1/K. Require the local sample cluster to fit in the ball where the bound is available.

Step 4 invokes a hypothesis "(iii)" that is not in the statement, and uses Lemma 4.1 as if it produced a homotopy of $M_{n,\delta'}$. It produces a homotopy of $M_{\delta'}$. Use the latter domain instead. The inclusion of M into $M_{\delta'}$ and the retraction T onto A imply that $T|_M$ is a homotopy equivalence; mod-2 top homology gives surjectivity. No Niyogi-Smale-Weinberger theorem is then needed.

Finally, $K \asymp 1/\tau$ does not follow from the assumptions of Theorem 4.21. Lemma 4.16 contains a term involving $\Lambda = \|\nabla II_M\|_\infty$ and obtains its reach-only simplification under an *additional* Λ bound. Also a lower bound on K is not an upper bound on K. Add $K \le C/\tau$, justified for example by a stated $\Lambda \le C/\tau^2$ hypothesis, if the final reach-only rate is intended.

Without that added simplification, retain the conditional bound CKh^2 with its actual K. Bounded reach by itself does not control derivatives of curvature: small-amplitude, high-frequency perturbations of a circle keep curvature and reach controlled while $\|\nabla II\|$ grows.

8. The PErr conclusion uses a larger domain than its proof

p. 21, Corollary 4.13 | Domain mismatch; normalization needs a hypothesis

The map estimate is stated and proved on $M_{n,\delta'}$, while the last paragraph says that the evaluation points Y_j lie in $M_{n,\delta}$. The smaller-domain estimate does not apply to every such Y_j.

Fix. Either restrict the evaluation points to the proved domain, or add an extension argument to the whole sample tube. Such an extension is available from the present constants; see the argument below. In all cases retain the displayed factor $1/\operatorname{tr} \hat{\Sigma}_n$, or assume a positive lower bound for that trace before replacing it by a constant. The covariance and its normalization should be defined in this paper.

An extension argument for the full sample tube

Proposed additional lemma; also clarifies the full-image assertion in Remark 4.15

For $x \in M_{n,\delta}$, put $w = T(x)$. If $d(x,M) < \delta'$, Lemma 4.1 already places [x,w] in the learned domain. Otherwise write $t = d(x,M) \ge 0.99\delta$, $p = \pi_M(x)$, $q = \pi_M(w)$, and choose a sample X_i with $|x - X_i| < \delta$. The containment term gives $a = |w - q| \le \delta/8$. Federer's chord inequality on M gives

$$|p - X_i|^2 \le \frac{\delta^2 - t^2}{1 - t/\tau} < (0.15\delta)^2,$$

$$|p - q|^2 \le \frac{(\delta + a)^2 - t^2}{1 - t/\tau} < (0.565\delta)^2.$$

Hence $|w - X_i| \le |w-q| + |q-p| + |p-X_i| < 0.84\delta$. Both endpoints of [x,w] lie in $B_\delta(X_i)$, so their whole segment lies there. The same ray argument therefore gives idempotence on the full sample tube. Its full image is the fixed-point set, which lies in the smaller tube. Applying the fiber estimate at w

and then along the fiber gives the O(h) map bound on the full learned domain. This extra lemma must precede the larger-domain use in PErr.

9. The degree arguments need mod-2 homology

p. 21, Theorem 4.7(iv); p. 20, Corollary 4.10; p. 25, Theorem 4.21 | Missing orientability assumption

M is not assumed orientable. For a closed connected nonorientable d-manifold, $H_d(M;\mathbb{Z}) = 0$, so an isomorphism on that group is not an integer-degree ±1 statement. Nonorientable embedded manifolds are allowed in higher codimension.

Fix. Use $H_d(M;\mathbb{Z}_2) = \mathbb{Z}_2$ for surjectivity. A homotopy equivalence induces a nonzero map on this group; a map missing a point would factor through a punctured manifold whose top mod-2 homology is zero. In Corollary 4.10, a local diffeomorphism between compact connected manifolds is a finite covering; since it is a homotopy equivalence, the induced map on π_1 is onto, so the covering has one sheet. Mod-2 degree alone would only say that the number of sheets is odd, so the π_1 argument is preferable.

Proposition 3.13 can also be shortened: use mod-2 top homology to detect the dimension directly. Its more elaborate integral-homology argument is not needed; this is a simplification rather than a counterexample to that proposition.

10. Keep covering-radius estimates deterministic until the last step

pp. 16-19, 21, 23-24; Proposition 4.3 and Theorem 4.7 | Logarithms and probability qualifiers are missing

For arbitrary deterministic samples there is no general relation $h \asymp n^{-1/d}$. Samples may cluster. For independent samples from a density bounded below on a compact smooth M, the standard high-probability upper bound has the form

$$h \le C\left(\frac{\log n + \log(1/\vartheta)}{n}\right)^{1/d}.$$

Thus the deterministic estimate $d_H(A,M) \le Ch^2/\delta$ gives an upper bound $C\delta^{-1}((\log n+\log(1/\vartheta))/n)^{2/d}$ on that event. Replace the displayed $\asymp n^{-2/d}/\delta$ statements by this qualified upper bound, or explicitly introduce logarithm-suppressing notation. Even the correct scale for h does not turn an *upper bound* on error into a two-sided asymptotic: $T = \pi_M$ is an exact minimizer with zero error.

The Vacher et al. lemma cited on p. 30 is stated for open Euclidean sets with an interior cone condition [2]. A manifold version follows from local volume bounds and a covering argument, but should be stated or proved. An r/2-net has at most $C_M r^{-d}$ points, each relevant ball has sampling mass at least cr^d, and the union bound gives $C_M r^{-d}\exp(-cnr^d)$. This yields the displayed bound for sufficiently large n.

Lemma 4.20 can be deterministic

pp. 23-24 | Simpler replacement; avoids a random-radius union-bound issue

Choose a regular d-simplex in T_zM at scale $r = C_dh$. Map its vertices by $\exp_z$ to M. For $r \le c_d\tau$, these points differ from their tangent-plane vertices by $O(r^2/\tau)$. By the definition of covering radius, choose a sample within h of each point. For C_d large and r/τ small, these are small perturbations of a regular simplex, have thickness at least θ_d, and lie in B(z,r) after adjusting constants. This works simultaneously for every z, for every h-dense sample.

The current proof makes the net and test balls depend on the random h, then applies a fixed-set independence/union-bound argument without controlling that dependence. The deterministic replacement removes the issue and the need for density and confidence parameters in the geometric lemma.

11. Interpolation and normalization need several corrections

pp. 30-32, Proposition 4.27 and Corollary 4.28 | Algebraic errors plus an unproved ambient norm estimate

Location	Correction
p. 30, sampling step	Use p = 2, q = ∞, regularity s-1, and derivative order 1 to obtain exponent s-2-d/2. The printed p = 1 gives a different exponent.
p. 30, prefactor	The printed factor δ^{-(d-D)/2} has the opposite sign from the mass factor δ^{-(D-d)/2} in Lemma 4.26. The proof of Theorem 4.23 repeats the same reversal with R.
p. 31, GN formula	If the higher norm is H^{s-2}, the coefficient is 1/2-(s-2)/D, not 1/2-s/D. The algebra on the next line already uses s-2.
pp. 31-32, regularity	The W^{2,s-2} statement needs at least s ≥ 4. An H^{s-2} bound cannot control two derivatives when s < 4.

More substantially, Lemma 4.26 only states a bound for π_A restricted to M. Corollary 4.28 cites it for a uniform H^{s-2} bound on the *ambient tube* $M_{n,R}$. This is not the same statement. Prove a separate extension estimate from the graph representation and normal coordinates before applying ambient Gagliardo-Nirenberg interpolation. Its constants must be uniform for the chosen domains; alternatively use a fixed smaller tube and a controlled extension operator.

Once that missing estimate is proved, the endpoint choices p = 2(s-2)/j and interpolation fraction j/(s-2), for j = 1,2, do yield the two printed rate exponents. The exponents themselves need not be discarded simply because their current justification is incomplete.

The claim on p. 32 that the constant is independent of D is not established. The slicing proof explicitly contains $\omega_{D-d}^{-1/2}$, the inverse square root of the unit normal-ball volume, in addition to chart and composition constants. Normalizing only by $\delta^{-(D-d)/2}$ does not remove this volume factor. Use normalized normal-fiber measure if dimension-independent constants are an intended goal, and track all remaining dependencies.

12. The final tangent and curvature claims need correction

p. 32, Corollary 4.29 | One false conclusion and missing regularity/identification

The last sentence claims that Proposition 4.3 gives an optimal tangent-space angle bound for k = 2. That proposition bounds the *fiber* direction, not the tangent space of A. Remark 4.4 explicitly explains why the inference is false. In its example the fibers equal $N_p M$ exactly while the normal graph can have arbitrarily steep tangent planes. Delete the sentence or add a uniform curvature/Hessian hypothesis and invoke a corrected Lemma 4.8.

For the second-derivative L∞ rate, apply the sampling inequality at derivative order 2 to a function of regularity s-1. To claim a decaying L∞ estimate via that route, one needs $s > 3+d/2$, stronger than the condition $s > 2+d/2$ stated for Lemma 4.26.

A difference of projection derivatives is also not literally a difference of second fundamental forms on different manifolds. Specify the identification of points (for example $\pi_M|_A$) and of tangent/normal spaces, then derive the geometric comparison. The displayed derivative estimates can support that calculation but do not themselves define it.

13. The fully discrete and approximate optimization claims have a different scope

pp. 2, 5-6 | Theorems must identify which objective and which minimizers they cover

Section 4 obtains pointwise rigidity from vanishing integrals over an entire sample tube. Replacing those integrals by a finite average over noised points only tests finitely many function values and derivatives. A smooth perturbation supported away from all tested points leaves the finite loss unchanged but can alter the map elsewhere, destroying the asserted global structure.

Fix. Call the Section 4 results results for the integrated, data-dependent objective. The informal “discrete main theorem” on p. 2 should use that scope. For the fully sampled objective, add a sampling/generalization argument with a controlled admissible class and a nonzero residual term. Likewise a quasi-minimizer with positive energy does not have the exact identities $T(X_i) = X_i$, $DT(T\text{-}Id) = 0$, or $T^2 = T$ used throughout these proofs; quantitative stability estimates would be a separate result.

The introduction also says that the image is M already without the orthogonality term. The current proof of Proposition 3.14 uses the segment property from Proposition 3.4, whose proof uses the E_4 displacement bound. Thus this stronger claim is not established by the supplied argument. Either give a proof without E_4 or narrow the introductory claim. The statements about unions of different dimensions and errors in the generated probability law similarly need precise theorems if they are to be claims of this paper.

14. Definitions, notation, and local proof corrections

Lower priority; worth fixing before another proof pass

Location	Issue and proposed correction
p. 1; asymptotic notes	log f ∈ Θ(log g) does not mean log f/log g → 1. Even a ratio tending to 1 is weaker than equality up to polylogarithmic factors. Define the desired convention explicitly; the second big-O condition in the suggested tilde-Θ definition should involve f, not g again.
p. 4, Def. 2.1	The W^{k,p} seminorm has a square instead of p, no pth root, and |α| = l instead of k. Set |f|_{W^{k,p}} = (Σ_{|α|=k} ‖D^αf‖_p^p)^{1/p}, with the usual modification for p = ∞.
p. 4, Rem. 2.2	The unweighted multiindex derivative norm is not exactly rotation invariant. For order 2, a 45-degree rotation changes Σ_{|α|=2}|D^α(x₁²)|² from 4 to 3. Use the full derivative tensor norm, or say the norms are equivalent. For p = 2, tensor weights are k!/α!.
pp. 2-3; notation	Use distinct symbols for codimension and Sobolev order. Define the covariance in PErr and use a consistent convention for tilde versus fully discrete objectives.
pp. 2, 4; sampling	vol_M is not automatically a probability measure. Normalize it, or explicitly sample from a probability density with respect to it.
p. 3; δ′	Equation (10) supplies a guaranteed inner radius, not the largest possible radius. Even solving its own quadratic bound in t gives a slightly larger admissible radius.
pp. 7-8; reach	Use closed solid tubes when taking reach. Open tubular neighborhoods are the integration domains; distinguish them from their closures in (7).
p. 9; Grönwall	The factor following |φ|′ ≤ C|φ|/(1-t) is (1-t)^{-C}, not generally (1-t)^{-1}. Since φ(0) = 0, the conclusion is unchanged.
pp. 12-13; regularity	A C² constant-rank map gives a C² image; it need not give a C∞ image. State the actual class when “smooth” might be read as C∞.
pp. 16-20; order	Establish the retraction/manifold/dimension conclusions on M_{δ′} before Proposition 4.3 invokes dim F = D-d. These structural conclusions do not require its quantitative estimate, so there is no need for a circular presentation.
p. 26; scaling	A change of units can set δ = 1, but the energy, estimator, Sobolev norm, and constants depend on the dimensionless geometry. State whether δ is fixed with n or is δ_n; the bound h²/δ retains its scale dependence.

A practical revision order

First keep the population objective with E_2 and narrow the introductory claims to what is proved. Next establish the finite-sample retraction and dimension statements on $M_{\delta'}$, then prove Proposition 4.3. Replace Proposition 4.9 by the short Taylor proof above and use mod-2 homology or covering-space arguments where appropriate. Make regularity selection and a smaller common projection tube explicit before Section 4.2. Rewrite the slice/graph lemma in normal-bundle charts, prove the missing ambient Sobolev estimate, and only then rederive the sampling and interpolation corollaries. Treat finite quadrature and positive-energy minimizers as separate extensions.

The strongest parts to retain are the C^1 rigidity-to-retraction mechanism, the quadratic tube gap $\eta = O(h^2/\delta)$, and the fixed-point manifold argument. The main work is to separate these unconditional zero-energy facts from statements requiring additional regularity.

External references checked

[1] H. Federer, Curvature Measures, Transactions of the AMS 93 (1959), 418-491. Theorem 4.18, printed p. 449, characterizes reach by normal chord deviation. Original paper.

[2] A. Vacher, B. Muzellec, A. Rudi, F. Bach, and F.-X. Vialard, A Dimension-free Computational Upper-bound for Smooth Optimal Transport Estimation, COLT 2021. Lemma 12, PDF p. 20, states the fill-distance estimate for a Euclidean interior-cone domain with density bounded below. Original paper.

All numbered draft locations refer to the supplied 32-page PDF. The counterexamples and replacement arguments in this review were checked directly; the numerical smoke checks are not used in place of the mathematical proofs.